# Calcurion Maths Worksheets

GCSE Mathematics

**Topic:** Area & Volume

Medium & Hard • 25 Questions

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## Section A – Medium (Q1–15)

**1.** Find the area of a rectangle with length 8 cm and width 5 cm. **[2]**

**2.** Find the area of a triangle with base 10 cm and height 6 cm. **[2]**

**3.** Find the area of a parallelogram with base 12 cm and height 7 cm. **[2]**

**4.** Find the circumference of a circle with radius 5 cm. Leave your answer in terms of $\pi$. **[2]**

**5.** Find the area of a circle with radius 4 cm. Leave your answer in terms of $\pi$. **[2]**

**6.** Find the volume of a cube with side length 6 cm. **[2]**

**7.** Find the volume of a cuboid with dimensions $5 \times 4 \times 3$ cm. **[2]**

**8.** Find the volume of a cylinder with radius 3 cm and height 10 cm. Leave your answer in terms of $\pi$. **[3]**

**9.** Find the area of a trapezium with parallel sides 6 cm and 10 cm and height 4 cm. **[3]**

**10.** Find the surface area of a cube with side 5 cm. **[2]**

**11.** Find the area of a sector of a circle with radius 6 cm and angle $60°$. Leave your answer in terms of $\pi$. **[3]**

**12.** Find the arc length of a sector with radius 8 cm and angle $45°$. Leave your answer in terms of $\pi$. **[3]**

**13.** Find the volume of a triangular prism with triangular base of area 12 cm$^2$ and length 9 cm. **[2]**

**14.** Find the area of a circle with diameter 14 cm. Use $\pi = 3.14$. **[3]**

**15.** Find the volume of a sphere with radius 3 cm. Leave your answer in terms of $\pi$. **[3]**

## Section B – Hard (Q16–25)

**16.** A cylinder has radius 4 cm and height 7 cm. Find its total surface area in terms of $\pi$. **[4]**

**17.** A cone has radius 3 cm and slant height 5 cm. Find its curved surface area in terms of $\pi$. **[3]**

**18.** Find the volume of a cone with radius 4 cm and height 9 cm. Leave your answer in terms of $\pi$. **[3]**

**19.** A sphere has volume $36\pi$ cm$^3$. Find its radius. **[3]**

**20.** A sector of a circle with radius 10 cm has angle $72°$. Find the area of the sector. Leave your answer in terms of $\pi$. **[3]**

**21.** A compound shape consists of a rectangle $8 \times 5$ with a semicircle of diameter 8 attached. Find the total area. Leave your answer in terms of $\pi$. **[4]**

**22.** A cylinder is hollowed out so that a smaller cylinder of radius 2 cm is removed from a larger cylinder of radius 5 cm and height 10 cm. Find the volume remaining in terms of $\pi$. **[4]**

**23.** Find the surface area of a closed cylinder with radius 3 cm and height 8 cm. Leave your answer in terms of $\pi$. **[4]**

**24.** A triangular prism has a right-angled triangular cross-section with legs 6 cm and 8 cm, and length 12 cm. Find its volume. **[4]**

**25.** A hemisphere has radius 6 cm. Find its total surface area (including the base) in terms of $\pi$. **[4]**

*End of Worksheet* • *Show all working clearly.*